MIT | Academy of Engineering (An Autonomous Institute Affiliated to Savitribai Phule Pune University)	ACTIVITY / ACTIVITIES	
Alandi (D), Pune - 412105	**ACADEMIC YEAR**	2026-2027
DEPARTMENT OF E&TC ENGINEERING	**SEMESTER**	VII
	CLASS	BTech
	DIVISION	E&TC

COURSE CODE	2307411T	**ACTIVITY NO.**	**2**
COURSE	Advanced Neural Network Architectures	**DATE**	**22-09-2025**

Activity 3: AI-Assisted Literature Review, Digital Mapping and Comparative Analysis of Sequential and Foundation Models

2306411T.CO.3: *Analyse sequential and foundation models in real-world applications. [L4]*

Course Activity Marks :20 Marks **Assessment Type: Internal Assessment Activity**

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1. Activity Objective

The objective of this activity is to enable students to:

- Select one real-world problem/application related to sequential or foundation models.
- Conduct an AI-assisted literature review of 10–15 research papers.
- Analyze the evolution of RNN → LSTM/GRU → Attention → Transformer → Foundation Models → LLMs.
- Compare architectures, datasets, methodologies, performance, advantages and limitations.
- Use AI tools for paper discovery, literature mapping, paper understanding and information extraction.
- Create digital diagrams and visual literature maps to represent the evolution and relationships among models.
- Identify research gaps and future research directions based on the reviewed literature.

Short CO-Based Description

Students will use AI-assisted literature-review tools to analyze 10–15 research papers on sequential and foundation models for a selected real-world application, create digital literature and architecture maps, compare research findings, and identify research gaps and future directions.

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2. Student Task

Each student/team shall select **ONE real-world problem statement/application**.

Suggested domains

- Healthcare, Finance, Agriculture, Education, Speech processing, NLP, Cybersecurity
- Transportation, Time-series forecasting, Generative AI, Healthcare NLP, Intelligent surveillance

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3. Literature Review Requirement

Students must select **10–15 relevant research papers** related to the selected problem.

Model Category	**Papers**
RNN/LSTM/GRU	2–3
Attention	2
Transformer	2–3
Foundation Models/LLMs	2–3
Recent review/comparative papers	1–2
Total	**10–15**

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4. AI-Assisted Literature Review

Students should use **at least three AI/research tools**.

Suggested tools

Paper Discovery

- Semantic Scholar, Google Scholar

Literature Mapping

- ResearchRabbit, Elicit

Paper Understanding

- ChatGPT, Gemini, Claude, NotebookLM, SciSpace

Citation Analysis

- Scite, Semantic Scholar

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5. Mandatory Digital Diagram Component

Students must **digitally draw diagrams** based on their literature findings.

The diagrams should not simply be downloaded from the Internet.

Students should **create their own visual representation** using tools such as:

Canva, draw.io / diagrams.net, Lucidchart, Figma, Mermaid , Miro , FigJam

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6. Digital Diagram 1 – Evolution of Models

Students must create an **Evolution Diagram** showing:

RNN ↓ LSTM / GRU ↓ Attention ↓ Transformer ↓ Pre-trained Models ↓ Foundation Models / LLMs

For every stage, students should show:

Architecture → Key Innovation → Limitation → Motivation for Next Model

Example structure

RNN *Sequential memory* ↓ **Limitation:** Vanishing gradient ↓ **LSTM** *Gated memory* ↓**Limitation:** Sequential computation ↓ **Attention** *Selective focus* ↓ **Limitation:** Computational complexity ↓ **Transformer** *Self-attention + parallel processing* ↓ **Foundation Models / LLMs**
Large-scale pretraining and generalization

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7. Digital Diagram 2 – Literature Mapping

Students must create a **digital literature map** for their 10–15 papers.

This diagram should visually show **how research has evolved for the selected application**.

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8. Digital Diagram 3 – Model-to-Application Mapping

Students should create a diagram connecting:

Model → Capability → Application → Advantage → Limitation

Example:

LSTM

→ Long-term temporal dependency
→ ECG analysis
→ Good sequential modeling
→ Higher sequential computation

Transformer

→ Long-range dependency + self-attention
→ ECG/clinical sequence analysis
→ Better global relationships
→ Higher computational requirement

LLM/Foundation Model

→ Large-scale pretrained representation
→ Clinical text/question answering
→ Transferability
→ Hallucination/domain adaptation challenges

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9. Digital Diagram 4 – Research Gap Map

This is an important **L4 analytical component**.

Students should identify gaps from the 10–15 papers and represent them digitally.

Example:

Literature Findings
↓
High accuracy
↓
But...
Gap 1: High computational cost
Gap 2: Limited real-time validation
Gap 3: Small datasets
Gap 4: Limited cross-domain validation
Gap 5: Limited interpretability
↓
Proposed Research Direction
Lightweight Transformer/LSTM Hybrid Model

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10. Literature Extraction Table

Students should prepare a structured table for all 10–15 papers.

Paper	Year	Application	Dataset	Model	Methodology	Performance	Limitation
P1				RNN			
P2				LSTM			
P3				GRU			
P4				Attention			
P5				Attention			

Paper	Year	Application	Dataset	Model	Methodology	Performance	Limitation
P6				Transformer			
P7				Transformer			
P8				Transformer			
P9				Foundation Model			
P10				LLM			
P11–15							

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11. Comparative Analysis

Students must compare:

Feature	RNN	LSTM	Attention	Transformer	Foundation/LLM
Sequential processing					
Long-term dependency					
Parallel processing					
Attention mechanism					
Scalability					
Computational cost					
Data requirement					
Application suitability					
Major advantage					
Major limitation					

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12. Critical Analysis

Students should answer:

1. Why was LSTM developed after RNN?
2. Why was Attention introduced?
3. What major limitation of RNN/LSTM was addressed by Transformer?
4. Why are Transformers suitable for large-scale pretraining?
5. Does a newer model always provide better performance?
6. Which model is most suitable for the selected application?
7. What are the computational limitations?
8. What research gaps are identified?
9. What future model/approach could address these gaps?

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13. AI Usage and Verification Log

Students must maintain an **AI Usage Log**.

AI Tool	Purpose	Example Query	Information Used	Verification
Semantic Scholar	Paper discovery			Original paper
ResearchRabbit	Literature mapping			References
Elicit	Paper extraction			Full paper

AI Tool	Purpose	Example Query	Information Used	Verification
ChatGPT	Concept understanding			Original paper/textbook
NotebookLM	Paper Q&A			Original PDF

Mandatory Rule

AI tools may assist the literature-review process, but students must read and verify the original research papers. AI-generated content cannot be submitted as a substitute for literature analysis.

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14. Final Student Deliverables

Each student/team should submit:

1. Literature Review Report

3. Digital Evolution Diagram

RNN → LSTM → Attention → Transformer → Foundation Model → LLM

4. Digital Literature Map

Showing relationships among the selected papers.

5. Digital Model–Application Mapping

6. Digital Research Gap Map

7. Critical Analysis

8. Future Research Proposal

9. AI Usage and Verification Log

10. References

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15. Suggested 20-Mark Rubric

Criteria	CO	5 Marks – Excellent	3 Marks – Good	1 Mark – Needs Improvement
1. Literature Search & AI-Assisted Review	3	10–15 highly relevant papers identified using appropriate AI/research tools; systematic search and verification demonstrated	Relevant papers identified with reasonable AI-tool usage; limited documentation	Few/irrelevant papers or inadequate verification
2. Technical & Comparative Analysis	3	Excellent analysis of RNN, LSTM, Attention, Transformer and Foundation/LLM models with evidence-based comparison	Good understanding and comparison with some missing analysis	Mainly descriptive; limited comparison
3. Digital Literature Mapping & Research Gap Analysis	3	Clear, technically accurate digital evolution map, literature map, application map and research-gap map	Diagrams are appropriate but analysis/mapping is partially developed	Incomplete, unclear or copied diagrams
4. Real-World Application, Critical Review	3	Strong application analysis, identifies meaningful research gaps, justifies	Reasonable application analysis and future scope	Weak application connection or

Criteria	CO	5 Marks – Excellent	3 Marks – Good	1 Mark – Needs Improvement
& Future Direction		conclusions and proposes a relevant future direction		unsupported conclusions

Total = 20 Marks
